Machine Learning

CS-871
Lecture-5

Instructor

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Graduate course for students of

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Non-Quadratic Regularizers

Regularizers

Regularizers and sensitivity

- we want to choose θ to achieve low empirical risk $\mathcal{L}(\theta)$
- but also, we'd like the predictor g_θ to not be too sensitive
- roughly: for x near $\tilde{x}$, $g_\theta(x)$ should be near $g_\theta(\tilde{x})$
  - sensitive predictors sometimes don't generalize well
  - insensitive predictors often generalize well
- a regularizer $r: \mathbb{R}^p \to \mathbb{R}$ is a function that measures the sensitivity of g_θ
- often predictor sensitivity corresponds to the size of θ
- another interpretation:
  - the regularizer encodes *prior information* we have about θ
  - specifically, that $r(\theta)$ is small
- with either interpretation, we want both $\mathcal{L}(\theta)$ and $r(\theta)$ small

Regularized empirical risk minimization

- in RERM we choose $\theta$to minimize $\mathcal{L}(\theta) + \lambda r(\theta)$
- $\lambda > 0$is the regularization hyper-parameter, used to trade off $\mathcal{L}(\theta)$and $r(\theta)$
- we choose λ(and r) by validation on a test set
- we use a regularizer to achieve better test set performance

Penalty based regularizers

- many common regularizers are given by a penalty function $q: \mathbb{R} \to \mathbb{R}$

$$r(\theta) = q(\theta_1) + \cdots + q(\theta_p)$$

- usually $q(a) \geq 0$for all a, and $q(0) = 0$
- $q(\theta_i)$expresses our displeasure in choosing predictor coefficient θ_i
- common examples:
  - sum square, quadratic, Tychonov, ℓ_2, or ridge regularizer:

$$q^{\text{sqr}}(a) = a^2, so r(\theta) = \| \theta \|_2^2$$

  - sum absolute, ℓ_1, or lasso regularizer:

$$q^{\text{abs}}(a) = | a |, so r(\theta) = \| \theta \|_1$$

  - nonnegative regularizer:

$$q^{\text{nn}}(a) = \begin{cases} 0, & a \geq 0 \\ \infty, & a < 0 \end{cases}$$

(requires predictor coefficients to be nonnegative)

Sensitivity of linear predictors

Feature perturbation

- consider a linear predictor $g_\theta(x) = \theta^T x$
- suppose the feature vector $x$changes to $\tilde{x} = x + \delta$
- $\delta$is the *perturbation* or change in x
  - we'll assume that any $\delta \in \Delta$is possible
- $\Delta$is called the *feature perturbation set*
- the change in prediction if $x$changes to $\tilde{x} = x + \delta$is

$$|\theta^T \tilde{x} - \theta^T x| = |\theta^T \delta|$$

- how big can this be, over all $\delta \in \Delta$?
- we define the *worst case sensitivity* as

$$\max_{\delta \in \Delta} |\theta^T \delta|$$

- it is evidently a measure of sensitivity

Worst case sensitivity with ℓ_2 perturbation

- let's take $\Delta = \{\delta \mid\mid \delta \mid\mid_2 \leq \epsilon\}$ (called an ℓ_2-ball)
- means the feature vector x can change to any $\tilde{x}$within $\ell_2$distance ϵ
- by Cauchy-Schwarz inequality,

$$| \theta^T \delta | \leq \| \theta \|_2 \quad \| \delta \|_2 \leq \epsilon \| \theta \|_2$$

- and the choice

$$\delta = \frac{\epsilon}{\| \theta \|_2} \theta$$

achieves this maximum change in prediction

- so the worst-case sensitivity is

$$\epsilon \| \theta \|_2$$

- justifies sum square regularizer

$$r(\theta) = \| \theta \|_2^2 = \theta_1^2 + \cdots + \theta_d^2$$

Worst case sensitivity with ℓ_∞ perturbation

- let's take $\Delta = \{\delta, |, |, \delta_i | \leq \epsilon \; i = 1 \ldots d\}$ (called an ℓ_∞-ball)
- also expressed as $\Delta = \{\delta \; || \; \delta \; ||_\infty \leq \epsilon\}$, Where, $|| \; \delta \; ||_\infty = \max_{i=1,\ldots,d} | \; \delta_i \; |$ is the ℓ_∞-norm of δ
- means any component of the feature vector $x$can change by up to ϵ
- how big can $| \; \theta^T \delta \; |$be, when $\delta \in \Delta$?
- the choice $\delta_i = \epsilon \, \text{sign}(\theta_i)$

maximizes the change in prediction, *i.e.,*

  - $\delta_i = \epsilon$if $\theta_i \geq 0$
  - $\delta_i = -\epsilon$if $\theta_i < 0$

- with this choice the change in prediction is $\epsilon |\theta^T \text{sign}(\theta)| = \epsilon(| \; \theta_1 \; | + \cdots + | \; \theta_d \; |) = \epsilon \; || \; \theta \; ||_1$
- so the worst case sensitivity is $\epsilon \; || \; \theta \; ||_1$
- justifies sum absolute regularizer $r(\theta) = || \; \theta \; ||_1 = | \; \theta_1 \; | + \cdots + | \; \theta_d \; |$

Ridge and lasso regression

- use square loss $\ell(\hat{y}, y) = (\hat{y} - y)^2$
- choosing θ to minimize $\mathcal{L}(\theta) + \lambda \parallel \theta \parallel_2^2$ is called *ridge regression*
- choosing θ to minimize $\mathcal{L}(\theta) + \lambda \parallel \theta \parallel_1$ is called *lasso regression*
- invented by (Stanford's) Rob Tibshirani, 1994
- widely used in advanced machine learning
- unlike ridge regression, there is no formula for the lasso parameter vector
- but we can efficiently compute it anyway (since it's convex)

Regulization with a constant feature

- suppose we have a constant feature $x_1 = 1$
- associated predictor coefficient θ_1 is the offset
  - since x_1 does not change, $\delta_1 = 0$ always
- so θ_1 does not contribute to predictor sensitivity
- for this reason it's common to *not* regularize the associated coefficient θ_1
- we modify sum square regularizer to

$$r(\theta) = \| \theta_{2:d} \|_2^2 = \theta_2^2 + \cdots + \theta_d^2$$

- we modify sum absolute regularizer to

$$r(\theta) = \| \theta_{2:d} \|_1 = | \theta_2 | + \cdots + | \theta_d |$$

Sparsifying regularizers

Sparse coefficient vector

- consider linear predictor $g_\theta(x) = \theta^T x$
- suppose $\theta$is sparse, *i.e.*, many of its entries are zero
- prediction $\theta^T x$does not depend on features $x_i$for which $\theta_i = 0$
- this means we select *some* features to use (*i.e.*, those with $\theta_i \neq 0$)
- (possible) practical benefits of sparse θ:
  - can improve performance when many regressors are actually irrelevant
  - makes predictor *simpler to interpret*
- choosing the sparsity pattern of θ (*i.e.*, which entries are zero) is sometimes called *feature selection*
- there are many ways to carry out feature selection

Sparse coefficient vectors via ℓ_1 regularization

using ℓ_1 regularization leads to sparse coefficient vectors

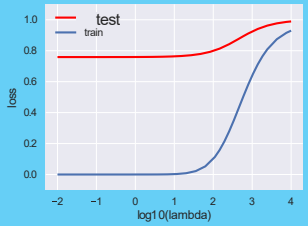
$$r(\theta) = \| \theta \|_1$$

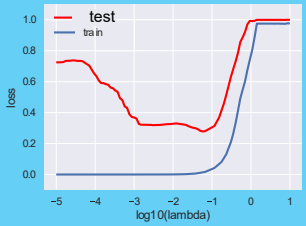
is called a *sparsifying regularizer*

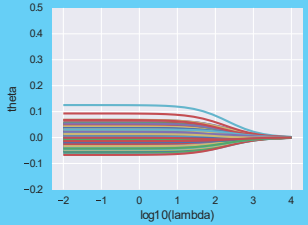
rough explanation:

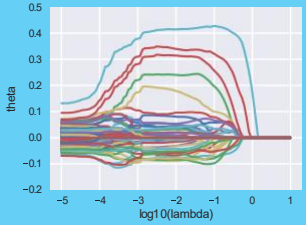
- for square penalty, once θ_i is small, θ_i^2 is very small
- so incentive for sum square regularizer to make a coefficient smaller decreases once it is small
- for absolute penalty, incentive to make θ_i smaller keeps up all the way until it's zero

Example



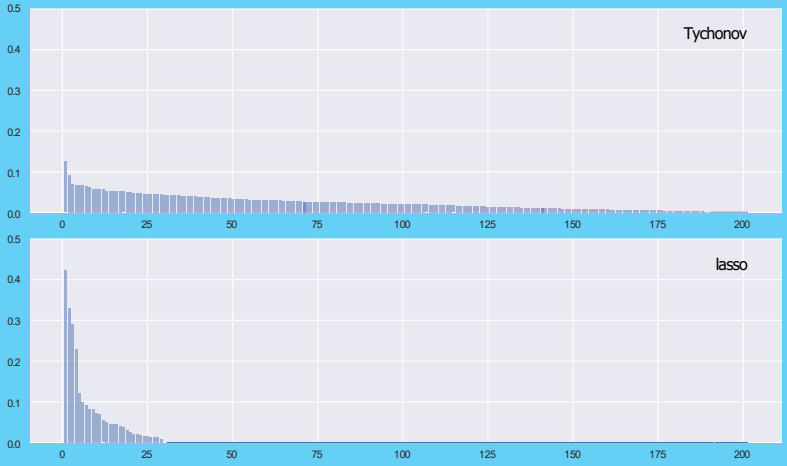






- artificially generated 50 data points, 200 features, only a few of which are relevant
- left hand plots use ridge regression, right hand use lasso

Example



► sorted | θ_i | at optimal λ

► lasso parameter has only 35 nonzero components; ridge regression has all 200 coefficients nonzero

Even stronger sparsifiers

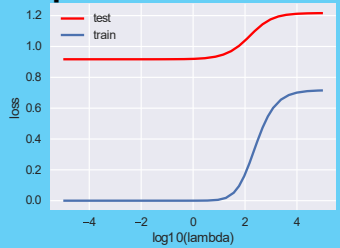
- $q(a) = |\ a\ |^{1/2}$
- called $\ell_{0.5}$regularizer
- but you shouldn't use this term since

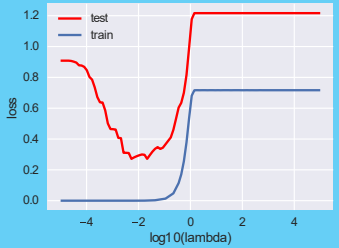
$$(|\ \theta_1\ |^{0.5} + \cdots + |\ \theta_d\ |^{0.5})^2$$

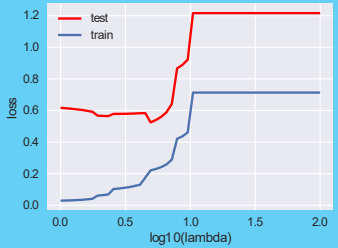
is not a norm (see VMLS)

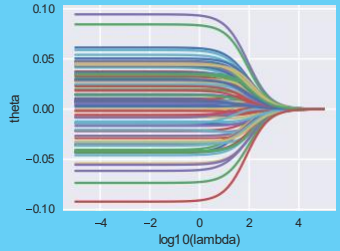
- 'stronger' sparsifier than ℓ_1
- but not convex so computing $\theta$is heuristic

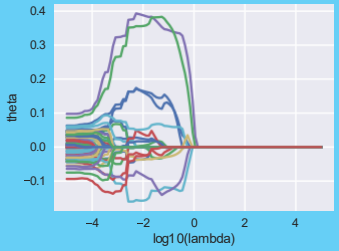
Example

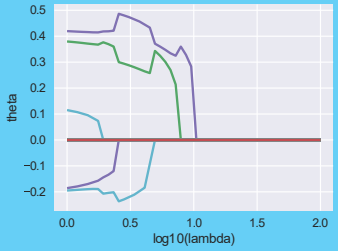












- ℓ_1 ,ℓ_2 and square root regularization

Nonnegative regularizer

Nonegative coefficients

- in some cases we know or require that $\theta_i \geq 0$
- this means that when $x_i$increases, so must our prediction
- we can think of this constraint as regularization with penalty function

$$q(a) = \begin{cases} 0, & a \geq 0 \\ \infty, & a < 0 \end{cases}$$

- example: $y$is lifespan, $x_i$measures healthy behavior i
- with quadratic loss, called *nonnegative least squares* (NNLS)
- common heuristic for nonnegative least squares: use $\left(\theta^{\mathrm{ls}}\right)_+$(works poorly)

Example



- feature vector $z == (1, u, (u - 0.2)_+, \ldots, (u - 0.8)_+)$
- nonnegative θ_i means both predictor function is convex (curves up) and nondecreasing
- NNLS loss 0.59, LS loss 0.30, heuristic loss 15.05

How to choose a regularizer

use out-of-sample or cross-validation to choose among regularizers

- for each candidate regularizer, choose A to minimize test error (and maybe a little larger . . .)
- use the regularizer that gives the best test error